

Final Exam

Macroeconomics II

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Problem 1 (Questions 1–20): Expectations, NKPC, shocks

- ① Definition. If $p_t \equiv \log P_t$, then inflation is _____
- (A) $\pi_t \equiv p_t - p_{t-1}$
 - (B) $\pi_t \equiv P_t - P_{t-1}$
 - (C) $\pi_t \equiv \log P_t$
 - (D) $\pi_t \equiv p_{t+1} - p_t$
- ② Three inflation objects. Which pair must be distinguished in NK models? _____
- (A) π_t and i_t
 - (B) π_t and $\mathbb{E}_t \pi_{t+1}$
 - (C) x_t and u_t
 - (D) β and κ
- ③ Adaptive expectations. Under adaptive expectations, we set _____
- (A) $\pi_t^e = \mathbb{E}_t \pi_{t+1}$
 - (B) $\pi_t^e = \pi_{t-1}$
 - (C) $\pi_t^e = \pi^*$ always
 - (D) $\pi_t^e = 0$ always
- ④ Forward-looking closure. A convenient NK choice of is _____

- (A) $\pi_t^e = \pi_{t-1}$
- (B) $\pi_t^e = \mathbb{E}_t \pi_{t+1}$
- (C) $\pi_t^e = \pi_t$
- (D) $\pi_t^e = i_t$

⑤ NKPC. The New Keynesian Phillips Curve is _____

- (A) $\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t$
- (B) $\pi_t = \pi_{t-1} + \kappa x_t + u_t$
- (C) $\pi_t = \kappa x_t + u_t$
- (D) $\pi_t = \beta \pi_{t-1} - \kappa x_t$

⑥ Expectations matter even if slack is zero. If $x_t = 0$ and $u_t = 0$, NKPC implies _____

- (A) $\pi_t = 0$ always
- (B) $\pi_t = \beta \mathbb{E}_t \pi_{t+1}$
- (C) $\pi_t = \pi_{t-1}$
- (D) $\pi_t = \kappa$

⑦ Meaning of x_t . In this course, the output gap is _____

- (A) $x_t \equiv y_t + y_t^{FE}$
- (B) $x_t \equiv y_t - y_t^{FE}$
- (C) $x_t \equiv \pi_t - \pi^*$
- (D) $x_t \equiv i_t - \pi_t$

⑧ Interpretation of β in NKPC. A larger β means _____

- (A) more forward-looking inflation (greater weight on expected future inflation)
- (B) steeper NKPC (larger response to x_t)
- (C) tougher monetary policy
- (D) larger cost-push shocks

⑨ Interpretation of κ in NKPC. A larger κ means _____

- (A) more weight on expected inflation
- (B) inflation responds more to slack x_t
- (C) the central bank reacts more to inflation
- (D) the natural real rate rises

⑩ Cost-push shock. In NKPC, u_t is best interpreted as _____

- (A) a demand shock moving the IS curve
- (B) a markup/energy/bottleneck shock shifting inflation for given x_t
- (C) the inflation target
- (D) the nominal interest rate

⑪ Calvo pricing. Each period, a fraction of firms that can reset prices is _____

- (A) θ
- (B) $1 - \theta$
- (C) β
- (D) κ

⑫ Calvo intuition. A firm that resets at time t chooses $P_t^\#$ to maximize profits while _____

- (A) the price remains fixed with some probability
- (B) money supply is constant
- (C) inflation is exactly zero
- (D) the output gap is zero

⑬ **Shift in NKPC.** Holding x_t fixed, an increase in $\mathbb{E}_t \pi_{t+1}$ _____

- (A) lowers π_t one-for-one
- (B) raises π_t by β times the change
- (C) does not affect π_t
- (D) changes κ

⑭ **Demand shock comovement.** A positive demand shock (raising x_t) tends to generate _____

- (A) $x_t \uparrow$ and $\pi_t \uparrow$
- (B) $x_t \downarrow$ and $\pi_t \uparrow$
- (C) $x_t \uparrow$ and $\pi_t \downarrow$
- (D) $x_t \downarrow$ and $\pi_t \downarrow$

⑮ **Cost-push comovement.** A positive cost-push shock $u_t > 0$ (with stabilization) tends to generate _____

- (A) $x_t \uparrow$ and $\pi_t \uparrow$
- (B) $x_t \downarrow$ and $\pi_t \uparrow$
- (C) $x_t \uparrow$ and $\pi_t \downarrow$
- (D) $x_t \downarrow$ and $\pi_t \downarrow$

⑯ **Calculation (NKPC).** Let $\beta = 0.98$, $\mathbb{E}_t \pi_{t+1} = 3\%$, $\kappa = 0.5$, $x_t = -2\%$, and $u_t = 0$. Then π_t is _____

- (A) 1.94%
- (B) 2.94%
- (C) 3.94%
- (D) 0.50%

⑪ Calculation (solve for slack). Suppose $\pi_t = 3\%$, $\beta = 1$, $\mathbb{E}_t\pi_{t+1} = 2\%$, $\kappa = 0.5$, and $u_t = 0$. Then x_t equals _____

- (A) 2%
- (B) 1%
- (C) 0%
- (D) -1%

⑫ Stabilizing a one-time cost-push shock. If $\mathbb{E}_t\pi_{t+1} = \pi^*$ and the central bank wants $\pi_t = \pi^*$ with $u_t = 1\%$ and $\kappa = 0.25$, then it must set _____

- (A) $x_t = -4\%$
- (B) $x_t = -1\%$
- (C) $x_t = +4\%$
- (D) $x_t = 0\%$

⑬ Anchored vs drifting. Expectations are drifting if _____

- (A) after a one-time shock, $\mathbb{E}_t\pi_{t+1}$ returns to π^*
- (B) after a one-time shock, $\mathbb{E}_t\pi_{t+1}$ stays above π^*
- (C) $x_t = 0$ always
- (D) $u_t = 0$ always

⑭ Key contrast. Relative to adaptive expectations, forward-looking expectations make inflation more sensitive to _____

- (A) last period inflation mechanically
 - (B) beliefs about future policy (credibility) today
 - (C) money demand only
 - (D) the money supply only
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Problem 2 (Questions 21–40): NK IS, Taylor rule, determinacy, ZLB

21) Fisher (ex-ante real rate). The expected real policy rate is approximately _____

- (A) $i_t + \mathbb{E}_t \pi_{t+1}$
- (B) $i_t - \mathbb{E}_t \pi_{t+1}$
- (C) $\pi_t - i_t$
- (D) $i_t - \pi_t$

22) NK IS (Euler) equation. A standard gap-form NK IS is _____

- (A) $x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma} (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n)$
- (B) $x_t = \mathbb{E}_t x_{t+1} + \frac{1}{\sigma} (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n)$
- (C) $x_t = -\sigma (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n)$
- (D) $x_t = \kappa (i_t - \mathbb{E}_t \pi_{t+1})$

23) Interpretation of σ . A larger σ implies the output gap responds _____

- (A) more strongly to the real-rate gap
- (B) less strongly to the real-rate gap
- (C) only to u_t
- (D) only to κ

- (24) Natural real rate. r_t^n is best interpreted as _____
- (A) the nominal policy rate
 - (B) the flexible-price real rate that would make $x_t = 0$
 - (C) the inflation target
 - (D) the cost-push shock
- (25) Calculation (NK IS). Suppose $\mathbb{E}_t x_{t+1} = 0$, $\sigma = 2$, $i_t = 4\%$, $\mathbb{E}_t \pi_{t+1} = 2\%$, and $r_t^n = 1\%$. Then x_t is _____
- (A) -0.5%
 - (B) -1%
 - (C) $+1\%$
 - (D) 0%
- (26) Calculation (real rate). If $i_t = 3\%$ and $\mathbb{E}_t \pi_{t+1} = 1.5\%$, then the expected real rate is _____
- (A) 4.5%
 - (B) 1.5%
 - (C) -1.5%
 - (D) 3%
- (27) Forward guidance channel. Holding i_t fixed, a higher $\mathbb{E}_t \pi_{t+1}$ implies _____
- (A) higher real rate and lower x_t
 - (B) lower real rate and higher x_t
 - (C) no change in x_t
 - (D) x_t must fall because expectations are higher
- (28) Taylor rule (general form). A standard rule in the NK core is _____

- (A) $i_t = \bar{i} + \phi_\pi(\pi_t - \pi^*) + \phi_x x_t$
- (B) $i_t = \pi_t - \phi_\pi x_t$
- (C) $i_t = r_t^n$ always
- (D) $i_t = \mathbb{E}_t \pi_{t+1}$

②9 Taylor principle in NK. A standard sufficient condition for determinacy is _____

- (A) $\phi_\pi < 1$
- (B) $\phi_\pi > 1$
- (C) $\phi_x < 0$
- (D) $\kappa = 0$

③0 Policy experiment sequence. After a one-time cost-push shock $u_t > 0$, the NK logic is _____

- (A) NKPC pushes $\pi_t \uparrow$; rule raises i_t ; NK IS yields $x_t \downarrow$; NKPC then helps bring π_t down
- (B) NKPC pushes $\pi_t \uparrow$; rule lowers i_t ; $x_t \uparrow$; inflation falls immediately
- (C) rule does nothing; inflation returns automatically
- (D) NK IS implies $x_t \uparrow$ when i_t rises

③1 ZLB constraint is _____

- (A) $r_t \geq 0$
- (B) $i_t \geq 0$
- (C) $\pi_t \geq 0$
- (D) $x_t \geq 0$

③2 ZLB calculation. A rule implies $i_t = -0.5\%$, but ZLB binds so $i_t = 0$. If $\mathbb{E}_t \pi_{t+1} = 1\%$, then the expected real rate is _____

- (A) -1.5%
- (B) -1%
- (C) 0%
- (D) $+1\%$

33 ZLB amplification. At the ZLB, a fall in expected inflation tends to _____

- (A) lower the real rate and raise demand
- (B) raise the real rate and lower demand
- (C) leave the real rate unchanged
- (D) eliminate the NKPC

34 Three-equation NK core consists of _____

- (A) IS (Euler), NKPC, Taylor rule
- (B) IS, LM, SRAS
- (C) AD, SRAS, LRAS
- (D) budget constraint, money demand, IS

35 Sacrifice ratio (simple). If expectations are anchored ($\mathbb{E}_t \pi_{t+1} = \pi^*$) and $u_t = 0$, then reducing inflation by 1 pp requires _____

- (A) $x_t = -1/\kappa$ pp
- (B) $x_t = -\kappa$ pp
- (C) $x_t = +1/\kappa$ pp
- (D) $x_t = 0$

36 Sacrifice ratio (numeric). If $\kappa = 0.2$ and you want inflation 0.5 pp lower (anchored expectations, $u_t = 0$), required x_t is _____

- (A) -2.5%
- (B) -0.1%
- (C) $+2.5\%$
- (D) -10%

37) 2-period equilibrium (calc). Suppose $\mathbb{E}_0\pi_1 = \pi^* = 2\%$, $\mathbb{E}_0x_1 = 0$, $\beta = 1$, $\kappa = 0.5$, $u_0 = 1\%$. Policy: $i_0 = r_0^n + \mathbb{E}_0\pi_1 + \phi_\pi(\pi_0 - \pi^*) + \phi_x x_0$ with $\sigma = 1$, $\phi_\pi = 2$, $\phi_x = 1$. Then π_0 is closest to _____

- (A) 2.0%
- (B) 2.33%
- (C) 2.67%
- (D) 3.0%

38) In the same 2-period equilibrium as previous question, x_0 is closest to _____

- (A) 0.0%
- (B) -0.33%
- (C) -0.67%
- (D) -2.0%

39) Determinacy intuition. The Taylor principle helps anchor inflation because it ensures that if π_t rises, policy _____

- (A) lowers the real rate, validating higher inflation
- (B) raises the real rate enough to create $x_t < 0$, pushing inflation down via NKPC
- (C) keeps the real rate fixed at r_t^n regardless of inflation
- (D) eliminates cost-push shocks

40) Old vs new Phillips curve. Relative to the accelerationist PC, the NKPC replaces _____

- (A) π_{t-1} with $\mathbb{E}_t \pi_{t+1}$
- (B) x_t with y_t
- (C) u_t with i_t
- (D) κ with β